Subrata Pal

su-pal.github.io
42 S University Pl, Apt. 02, Stillwater, OK–74075
subrata-pal1
supal@okstate.edu
Google Scholar
subratapal.1793@gmail.com
0000-0002-8070-7318
+1 (405) 762-8690

Education

Ph.D. Candidate in Organic Chemistry Aug. 2022 – Present (Expected Dec. 2027)
Department of Chemistry, Oklahoma State University *Stillwater, OK*

- Advisor: Dr. Spencer P. Pitre
- Dissertation: *"Structure-Activity Relationship Investigation of Cobalt Complexes for Phoctocatalyzed Reactions Under Visible Light Irradiation."* (working title)

MS in Organic Chemistry Jan. 2016 – Apr. 2018
Department of Chemistry, Shahjalal University of Science & Technology *Sylhet, Bangladesh*

- Advisor: Dr. Mohammad Mizanur Rahman Khan
- Dissertation: *"Synthesis and Investigation of PVA-ZnS and PVA-AgI-ZnO Nanocomposite Films: Towards Efficient and Stable Photocatalyst for Organic Dye Removal"*

B.Sc. in Chemistry Jan. 2011 – Dec. 2015
Department of Chemistry, Shahjalal University of Science & Technology *Sylhet, Bangladesh*

Research Experience

Graduate Research Assistant Jan. 2025 – Present
Pitre Lab, Oklahoma State University *Stillwater, OK*

- Synthesized a library of cobaloxime catalysts and evaluated their catalytic performance alongside commercially available Co(II) catalysts in Giese-type reactions with diverse Michael acceptors and radical precursors to establish structure–activity relationships and identify viable alternatives to Vitamin B_{12} catalysts.
- Enabled Acid-Free Regioselective Cobalt-Photocatalyzed C–4 Minisci Alkylation through Urea Activation of Pyridines.
- Employed TLC, 1D/2D NMR spectroscopy, and GC–MS to characterize reaction intermediates and products and elucidate catalytic mechanisms.
- Performed computational studies using Gaussian and ORCA for geometry optimization and frequency calculations, followed by energetic evaluations to calculate bond dissociation energies (BDEs), redox potentials, and nucleophilicity of cobalt complexes.
- Mentored one undergraduate researchers.

Graduate Research Assistant Jan. 2016 – Apr. 2018
MMR Khan Research Group, Shahjalal University of Science & Technology *Sylhet, Bangladesh*

- Synthesized PVA-ZnS and PVA-AgI-ZnO nanocomposite films with varied PVA compositions and investigated their photocatalytic activity toward organic dye degradation.
- Employed FT–IR, UV-vis and PL spectroscopy to characterize the nanocomposite films.

Publications

*(‡ indicates equal contribution & * indicates corresponding author)*

2. **Pal, S.**‡; Pauze, M.‡; Shafiei, N.‡; Pitre, S. P.* "Base Metal Photocatalysts: A Bright Future in Photoredox Catalysis?" *Organometallics* **2026**, *45 (6)*, 587 – 612.

1. Khan, M. M. R.*; **Pal, S.**; Hoque, M. M.; Alam, M. R.; Younus, M.; Kobayashi, H. "Simple Fabrication of PVA–ZnS Composite Films with Superior Photocatalytic Performance: Enhanced Luminescence Property, Morphology, and Thermal Stability." *ACS Omega* **2019**, *4 (4)*, 6144 — 6153.

Conference Presentations

- *67th Annual ACS Pentasectional Meeting*. (Apr. 2026) **Pal, S.**; Pauze, M.; Pitre, S. P. "Structure-Activity Relationship Investigation of Cobalt Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Oral Presentation, Pittsburg State University, Pittsburg, Kansas, USA.
- *66th Annual ACS Pentasectional Meeting*. (Mar. 2025) **Pal, S.**; Pitre, S. P. "Investigation of Cobalt(II) Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Oral Presentation, Oral Roberts University, Tulsa, Oklahoma, USA.
- *ACS Southwest Regional Meeting*. (Oct. 2024) **Pal, S.**; Pitre, S. P. "Investigation of Cobalt(II) Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Poster Presentation, Waco Convention Center, Waco, Texas, USA.
- *ACS Southwest Regional Meeting*. (Nov. 2023) **Pal, S.**; Pitre, S. P. "Cobaloxime-Photocatalyzed Minisci Reactions Under Visible Light Irradiation", Poster Presentation, Omni Oklahoma City Hotel, Oklahoma City, Oklahoma, USA.
- *International Conference on Nanotechnology and Condensed Matter Physics*. (Jan. 2018) **Pal, S.**; Khan, M. M. R.; Oliullah, M. "Synthesis, Interaction and Thermal Properties of PVA-ZnS Nanocomposite Films: Application as Stable Photocatalyst for Organic Dye Removal", Poster Presentation, Dept. of Physics, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.

Awards & Fellowship

- **National Science & Technology (NST) Fellowship**, *Ministry of Science and Technology, Bangladesh* (2017)

Teaching Experience

Graduate Teaching Assistant — Aug. 2022 – Dec. 2024
Department of Chemistry, Oklahoma State University — *Stillwater, OK*

- Instructed Courses: General Chemistry I Lab, Organic Chemistry I, Survey of Organic Chemistry and Lab.
- Led laboratory sessions of approx. 50 undergraduate students per semester, graded lab reports and exams.

Lecturer (Chemistry) — Sep. 2019 – Jul. 2022
Green University of Bangladesh (GUB) — *Dhaka, Bangladesh*

- Instructed Courses: General Chemistry, Organic Chemistry (Both Theory & Laboratory).
- Led weekly discussion sections and laboratory sessions for approximately 200 undergraduate students per semester.

Laboratory & Technical Skills

- **Synthetic Techniques:** Air- and moisture-sensitive (Schlenk/glovebox) techniques, organometallic synthesis, multi-step organic synthesis, photochemical reactions
- **Instrumentation:** NMR (1D/2D), UV–Vis, TLC, GC–MS, FT–IR, Wet chemistry including extractions, distillations, standard/solution preparation
- **Software:** MestReNova, ChemDraw, OriginPro, MS Office Suite, Gaussian, Avogadro, Molden, LaTeX

Professional Affiliations

- **American Chemical Society (ACS)**, *Student Member* since Oct. 2025
- **ACS Graduate Student Organization–OSU (ACSGSO–OSU)**, *Secretary* since Sep. 2025
- **ChemFusion**, *Advisor* since Mar. 2021

References

Dr. Spencer P. Pitre — spencer.p.pitre@okstate.edu
Associate Professor, Research Advisor — +1 (405) 744-2483